

Notes: Titman (JFE, 1984)

The Effect of Capital Structure on a Firm's Liquidation Decision

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1 Big picture

- **Question.** Can capital structure affect real decisions even when direct bankruptcy costs are small? Titman answers yes: debt changes the firm's future liquidation incentives, and those incentives matter because liquidation imposes costs on customers, workers, and suppliers.
- **Core idea.** The firm is a nexus of implicit contracts. Customers who buy durable goods, workers with firm-specific human capital, and suppliers with relationship-specific investments may be harmed if the firm liquidates or reorganizes.
- **Time-inconsistency problem.** Ex ante, the firm wants to promise not to liquidate unless the liquidation value exceeds the operating value by enough to compensate outsiders for their loss. Ex post, after customers and suppliers have sunk relationship-specific investments, equityholders may want to liquidate whenever liquidation value merely exceeds operating value.
- **Capital-structure mechanism.** Debt and preferred stock can pre-position or bond the firm. By choosing claims with different control rights and payoff priorities, the firm can commit to a liquidation policy closer to the ex ante efficient policy.
- **Product-market implication.** Customers rationally discount the price they are willing to pay for products sold by a firm that may liquidate. A firm with high liquidation probability receives worse terms of trade in its product market.
- **Debt cost.** More debt increases the probability of bankruptcy and therefore increases the probability of inefficient liquidation. This lower expected product-market value is an indirect cost of debt.
- **Security-design implication.** The optimal financing mix may include debt, equity, and possibly preferred stock because different securities allocate control and liquidation incentives across states.
- **Testable implication.** Firms that impose high liquidation costs on customers, workers, or suppliers should use less debt. Durable-goods producers, firms selling products requiring future service, and firms relying heavily on relationship-specific investments should have lower leverage.

2 Incremental contribution of the paper

2.1 Contribution relative to Modigliani-Miller and tax-bankruptcy tradeoff theories

- Early capital-structure theory emphasized taxes and direct bankruptcy costs.
- Titman adds a new indirect cost of debt: debt can increase the probability of liquidation, and liquidation can destroy value for nonfinancial stakeholders.
- This cost is borne by the firm ex ante because customers, suppliers, and workers adjust prices, wages, and contract terms in anticipation of future liquidation risk.

Interpretation: The paper shifts the capital-structure debate away from court costs and toward real effects in product and input markets.

2.2 Contribution relative to agency and incomplete-contract theories

- Jensen and Meckling-style agency theory studies conflicts between managers, shareholders, and bondholders.
- Titman expands the agency relationship to include customers, workers, and suppliers as principals affected by the firm's future liquidation decision.
- The firm's capital structure becomes a bonding mechanism for implicit contracts with outsiders, not merely a contract among financial claimants.

Interpretation: This is a key conceptual extension: financial policy supports nonfinancial contracting relationships.

2.3 Contribution relative to bankruptcy-cost critiques

- Warner and Haugen-Senbet question whether conventional bankruptcy costs are large enough to explain capital structure.
- Titman shows that liquidation costs need not be legal bankruptcy costs. They can arise because customers lose service, workers lose firm-specific capital, or suppliers lose relationship-specific value.
- These costs can matter even when formal bankruptcy costs are low.

Interpretation: The relevant cost of financial distress is not just what lawyers and courts consume. It is also the loss imposed on counterparties and reflected in ex ante contracting terms.

2.4 Contribution to product-market finance

- The paper links corporate financial structure to product-market competition and customer willingness to pay.
- The model explains why consumers might avoid purchasing durable goods from a financially distressed producer.
- It predicts that financing choices affect sales prices, demand, and product-market reputation.

Interpretation: This makes Titman an early foundational paper in the literature on interactions between financial policy and product-market strategy.

2.5 Contribution to security design

- The paper shows that the mix of debt, preferred stock, and equity affects who controls the liquidation decision in different states.
- Preferred stock can be useful because it changes equityholders' liquidation incentives without necessarily triggering bankruptcy.
- Capital structure can partly replicate state-contingent contracts that would otherwise be costly to write and enforce.

Interpretation: Financing instruments are not just cash-flow claims. They are commitment devices that shape future real decisions.

3 Model environment

3.1 Agents and timing

The model has two periods.

- In period 0, the firm chooses production M and a financial structure.
- The firm produces machines using $K(M)$ units of the numeraire good, where $K(M)$ is increasing and strictly convex.
- Machines are sold to customers at the end of period 0.
- In period 1, one of N states of nature θ_i is realized.

- The firm then either continues operating or liquidates.

Interpretation: The key feature is that customers buy machines before they know whether the firm will be around to maintain them.

3.2 Operating and liquidation values

In state θ_i , the firm has:

- operating value $V_1(\theta_i)$;
- liquidation value $\delta(\theta_i)K(M)$;
- customer liquidation cost $C(\theta_i)$.

The liquidation policy is

$$\mu(\theta_i) = \begin{cases} 1, & \text{if the firm liquidates in state } \theta_i, \\ 0, & \text{if the firm continues in state } \theta_i. \end{cases} \quad (1)$$

Interpretation: Liquidation value is what financial claimants receive from selling assets. Customer cost is the external loss imposed on buyers of the firm's machines.

3.3 Customer valuation and durable goods

Customers care about the total cost of machine services. If the producer continues, maintenance is cheaper. If the producer liquidates, the machine becomes more costly to operate because specialized maintenance or parts are harder to obtain.

Let the machine value in state θ_i be

$$P[0, \theta_i] \quad \text{if the firm does not liquidate,} \quad (2)$$

$$P[1, \theta_i] \quad \text{if the firm liquidates.} \quad (3)$$

The difference is the cost imposed on customers:

$$P[0, \theta_i] - P[1, \theta_i] = C(\theta_i)/M. \quad (4)$$

Interpretation: The firm bears expected customer liquidation costs through a lower period-0 machine price.

3.4 Period-0 firm value

The period-0 machine price reflects the firm's liquidation policy:

$$P_0(\mu) = \sum_{i=1}^N q_i P[\mu(\theta_i), \theta_i], \quad (5)$$

where q_i is the state price of the numeraire good in state θ_i .

Period-0 firm value is

$$P_0(\mu)M - K(M) + \sum_{i=1}^N q_i [\mu(\theta_i)\delta(\theta_i)K(M) + (1 - \mu(\theta_i))V_1(\theta_i)]. \quad (6)$$

Interpretation: Period-0 value includes product-market revenues, production cost, and the discounted value of future operating or liquidation payoffs.

4 Optimal liquidation and time inconsistency

4.1 Ex ante optimal liquidation rule

The ex ante value-maximizing firm liquidates in state θ_i only if

$$\delta(\theta_i)K(M^*) - C(\theta_i) > V_1(\theta_i). \quad (7)$$

Interpretation: The firm liquidates only if the liquidation value exceeds the operating value by more than the customer cost. This rule maximizes total ex ante value because customers anticipate and price liquidation risk.

4.2 Ex post liquidation rule

After customers have bought the machines, an unconstrained firm ignores the customer loss $C(\theta_i)$ and liquidates whenever

$$\delta(\theta_i)K(M^*) > V_1(\theta_i). \quad (8)$$

Interpretation: This is the time-inconsistency problem. Ex ante, the firm wants to internalize customer losses. Ex post, once customers are locked in, the firm wants to ignore those losses.

4.3 Why explicit contracts may be costly

The firm could solve the problem by writing state-contingent contracts with customers or by specifying liquidation penalties. But such contracts may be costly because:

- states of nature are hard to describe;
- liquidation-related customer losses are hard to verify;
- enforcement and litigation are costly;
- contracts with many customers, workers, and suppliers are cumbersome.

Interpretation: Capital structure is attractive because it can partly implement the desired policy with fewer explicit contracts.

contracts. The concluding section summarizes the work and discusses the testable implications of the theory.

2. The time-inconsistency of the firm's optimal liquidation policy

In the model which follows, the firm can be viewed from the perspective suggested by Alchian and Demsetz (1972) and Jensen and Meckling (1976); as a nexus for contracting relationships between the firm's different classes of security holders and its customers, workers and suppliers. However, firm value is defined in the more traditional manner as the summed values of its outstanding securities. The analysis which follows demonstrates that a value-maximizing firm's optimal liquidation policy is not, in general, time-consistent. The liquidation policy which maximizes the firm's value in period 0 specifies that the firm should act to maximize the aggregate wealth of all of its associates (including its customers, workers and suppliers) in period 1 rather than just the wealth of its security holders. These objectives are inconsistent whenever any of these outside associates suffer a cost from the firm's liquidation. Although

Original paper excerpt: model setup and optimal liquidation rule.

5 Capital structure as pre-positioning

5.1 Debt and control rights

Debt changes the liquidation decision by transferring control to bondholders in bankruptcy states. Bondholders have a high-priority claim to liquidation proceeds, so they are more willing to liquidate than equityholders.

Equityholders control the firm when it is not bankrupt. Bondholders effectively control liquidation when bankruptcy occurs. Therefore, the debt level determines which states of nature are bankruptcy states.

Interpretation: The firm's financial structure becomes a state-contingent control allocation device.

5.2 Proposition 1: sufficient conditions for implementation

Proposition 1 gives sufficient conditions under which capital structure implements the ex ante optimal liquidation policy:

1. The firm is bankrupt in all and only those states in which liquidation is ex ante optimal:

$$V_1(\theta_i) < \delta(\theta_i)K - C(\theta_i). \quad (9)$$

2. The debt claim is at least the customer-adjusted liquidation value whenever the firm is bankrupt:

$$D \geq \delta(\theta_i)K - C(\theta_i). \quad (10)$$

3. Debt plus preferred stock is at least the gross liquidation value in all states:

$$P^f + D \geq \delta(\theta_i)K, \quad \forall \theta_i. \quad (11)$$

Interpretation: These conditions align financial-claimant incentives with the ex ante efficient liquidation rule.

5.3 Proposition 2: simple capital structure under constant costs

If δK and C are constant across states and bankruptcy occurs whenever operating value falls below debt obligations, the optimal policy can be implemented if the firm:

1. issues short-term debt maturing in period 1 with face value

$$D = \delta K - C; \quad (12)$$

2. issues preferred stock satisfying

$$P^f \geq C. \quad (13)$$

Interpretation: A simple debt-plus-preferred-stock structure can implement a state-contingent liquidation policy without writing direct state-contingent contracts with customers.

Proposition 1. A firm will liquidate according to its optimal policy if its capital structure contracts are chosen so that

- (1) *it is bankrupt in all those states of nature and only those states of nature in which*

$$V_1(\theta_i) < \delta(\theta_i)K - C(\theta_i),$$

- (2) *$D \geq \delta(\theta_i)K - C(\theta_i)$ whenever the firm is bankrupt, and*
- (3) *$P^f + D \geq \delta(\theta_i)K, \forall \theta_i$.*

Proposition 1: sufficient conditions for capital-structure implementation.

Proposition 2. If δK and C are constant across states of nature and bankruptcy occurs whenever the firm's operating value falls below its debt obligations, the firm will liquidate according to its optimal policy if it

- (a) *issues short term debt which matures in period 1 with a maturity value equal to $\delta K - C$, and*
- (2) *issues preferred stock in an amount where $P^f \geq C$.*

Proposition 2: simple implementation with debt and preferred stock.

6 Figures and graphical intuition

6.1 Figure 1: when a simple debt level cannot implement the optimal policy

Read:

- The solid curve is operating value $V_1(\theta_i)$.
- The dashed curve is liquidation value net of customer costs, $\delta(\theta_i)K(M^*) - C(\theta_i)$.
- The two curves cross twice.
- No single state-independent debt level can make bankruptcy occur exactly in the states where liquidation is ex ante optimal.

Economic message: When the optimal liquidation region is nonmonotone in operating value, simple debt cannot exactly implement the optimal liquidation rule. More complex debt maturity, priority, or covenants may be needed.

6.2 Figure 2: when a simple debt level can implement the optimal policy

Read:

- The curves cross only once.
- A debt face value D^* equal to operating value at the intersection can make bankruptcy occur exactly when operating value is below the threshold.
- In this case, bankruptcy control aligns with the ex ante optimal liquidation policy.

Economic message: If the optimal liquidation rule is monotone in operating value, a simple debt threshold can implement the policy.

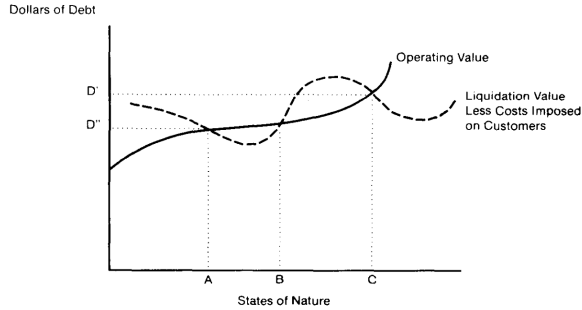


Fig. 1. The operating value of the firm ($V_1(\theta_i)$) and its liquidation value less the costs imposed on its customers [$\delta(\theta_i)K(M^*) - C(\theta_i)$] are plotted against the state of nature θ_i , where θ_i is ordered so that $V_1(\theta_i)$ is a monotonically increasing function. This figure illustrates the case where these two values intersect twice, implying that for the relatively low operating values in the region AB , $V_1(\theta_i) > \delta(\theta_i)K(M^*) - C(\theta_i)$, and for the higher operating values in the region BC , $V_1(\theta_i) < \delta(\theta_i)K(M^*) - C(\theta_i)$. Since $V_1(\theta_i) > V_1(\theta_j)$, the firm cannot implement the optimal liquidation policy by selecting a debt level D^* and specifying that the firm is bankrupt whenever $V_1(\theta_i) < D^*$.

Figure 1: simple debt cannot implement a nonmonotone optimal liquidation region.

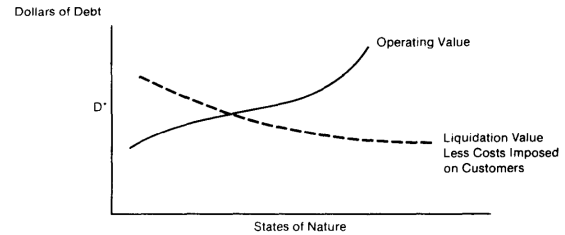


Fig. 2. As in fig. 1, the operating value of the firm ($V_1(\theta_i)$) and its liquidation value less the costs imposed on its customers [$\delta(\theta_i)K(M^*) - C(\theta_i)$] are plotted against the state of nature θ_i , where θ_i is ordered so that $V_1(\theta_i)$ is a monotonically increasing function. This figure illustrates the case where these two values intersect only once. In this case the firm can implement its optimal liquidation policy by choosing a debt level D^* which equals the firm's operating value at the point of intersection and specifying that the firm is bankrupt whenever $V_1(\theta_i) < D^*$.

Figure 2: simple debt can implement a monotone liquidation threshold.

7 Empirical and economic implications

7.1 Leverage predictions

The theory predicts lower leverage for firms that impose larger liquidation costs on nonfinancial stakeholders. Examples include:

- durable-goods producers whose products require future service or spare parts;

- firms selling specialized equipment;
- firms relying on long-term supplier relationships;
- firms with workers who accumulate firm-specific human capital;
- firms whose customers make relationship-specific investments.

Interpretation: High liquidation costs make debt more expensive because debt increases the probability that liquidation control passes to bondholders.

7.2 Product-market predictions

The model predicts that customers rationally respond to financial distress:

- financially weak firms may have to charge lower prices;
- customers may avoid purchasing durable goods from highly levered firms;
- product-market reputation can depend on financial policy;
- firms may choose conservative leverage to reassure customers.

Interpretation: This is an early statement of the product-market cost of financial distress.

7.3 Security-design predictions

The model suggests that different securities can support different commitments:

- short-term debt can define bankruptcy-trigger states;
- preferred stock can discipline equityholders' incentives without necessarily transferring control to bondholders;
- covenants, seniority, and maturity can refine the state-contingency of control.

Interpretation: Capital structure is a mechanism for allocating control rights across states, not merely a way to split cash flows.

8 Short summary

- Liquidation can harm customers, suppliers, and workers.
- Customers anticipate liquidation risk and pay less for products of firms that are more likely to liquidate.
- The ex ante optimal liquidation policy internalizes these customer costs.
- The ex post liquidation policy of unconstrained security holders does not internalize these costs.
- This creates a time-inconsistency problem.
- Capital structure can pre-position the firm to implement a better liquidation policy.
- Debt affects when the firm enters bankruptcy and when bondholders control liquidation.
- Preferred stock can help align equityholder incentives.
- Firms with high stakeholder liquidation costs should use less debt.

- The paper is an early foundation for theories linking financial policy to product-market relationships and stakeholder contracting.

9 Conclusion

9.1 Core conclusion

Titman shows that capital structure affects firm value because it affects future real decisions. Debt can make liquidation more likely. If liquidation harms customers, workers, or suppliers, those harms are priced into the firm's current transactions. The firm can therefore use capital structure as a commitment device to implement a more efficient liquidation policy.

9.2 Economic intuition

The firm wants to promise its customers that it will not liquidate opportunistically after they buy its products. Direct promises may be costly to write or enforce. By choosing the right mix of debt, equity, and preferred stock, the firm can allocate control rights so that liquidation occurs in the right states. Financial structure thereby supports implicit product-market contracts.

9.3 Takeaway

The paper's lasting lesson is that capital structure is not only about taxes, bankruptcy court costs, or conflicts among financial claimants. It also governs real commitments to nonfinancial stakeholders. Debt policy can shape customer trust, product prices, and the firm's ability to maintain valuable long-term relationships.